# Thermodynamics Clements Invitational: Key

Clements High School

12/16/17

## 1 Thermodynamics

1.1. C

1.2. C

1.3. D

1.4. B

1.5. B

1.6. C

1.7. C

1.8. A

1.9. B

1.10. B

1.11. D

1.12. A

1.13. D

1.14. C

1.15. -40

1.16. -17.78 (accept -17 to -18)

1.17. 4.44 (accept 4 to 5)

1.18. 0

1.19. 100

1.20. -3.9 (accept -3 to -4 if work is correct)

1.21. E

1.22. C

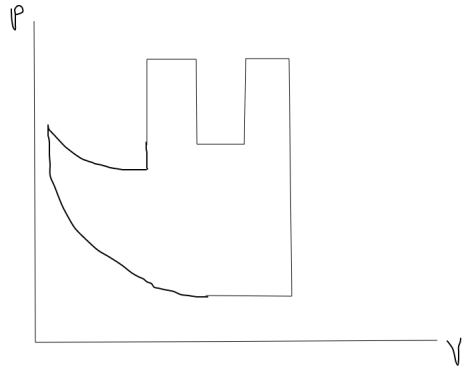
1.23. A

1.24. D

1.25. B

## 2 A really cool engine!!

(5 points) Relative location of points should be accurate, distances need not be correct.




GRADE WITH LENIENCY-± 5 units for each answer is acceptable in first table and ± 1% of answer is acceptable in second table (except for exact 0 answers)

In the following table, let 1 denote the initial state of the system, and 10 denote the last state of the system before it returns to initial conditions. Fill in all missing variables. 1 point per blank

| Variables | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| P(atm) | 100.00 | 100.00 | 70 | 70 | 100.00 | 100.00 | 10 | 10 | 75.00 | 45 |
| V(L) | .500 | 1.00 | 1.00 | 1.500 | 1.500 | 2.00 | 2.00 | 1.00 | .3 | .500 |
| T(K) | 609.3 | 1218.7 | 853.06 | 1279.59 | 1827.99 | 2437.3 | 243.73 | 121.87 | 272.827 | 272.827 |

Calculate the amount of work done by the system in each process, as well as the amount of heat added or released in each process. 1 point per blank

| Step | Work(J) | Heat(J) |
|---|---|---|
| 1 | 5066 | 12622 |
| 2 | 0 | -4557 |
| 3 | 3546 | 8865 |
| 4 | 0 | 6880 |
| 5 | 5066 | 12699 |
| 6 | 0 | -27356 |
| 7 | -1013 | -2527 |
| 8 | -1882 | 0 |
| 9 | 1165 | 1165 |
| 10 | 0 | 4237 |

Efficiency $\eta = 25\%$

# Clements Invitational - Repaired practice edition - Answer key

### Division C | Thermodynamics | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

### ANSWER KEY - Review only after submitting.

**1-13. (1 points) Authored replacement for source item Question 1-13: What is the SI unit of heat energy needed to melt a sample? Distinguish it from the unit of specific latent heat of fusion.**

answer: Heat energy is in joules (J). Specific latent heat of fusion is energy per mass, J/kg. The original choices named concepts rather than units. (1 points).

**1-20. (6 points) Authored replacement for source item Question 1-20: Solve this case and explain your reasoning: A system receives 500 J heat and does 200 J work. Find delta U and chemistry work.**

reasoning: Q=+500 J, Wby=+200 J.
delta U=500-200=300 J.
won=-200 J, giving q+won=300 J. (3 points).

conclusion: Both sign conventions give the same energy change. (3 points).